1 Spring–mass models
- Announcements
- IVPs for linear models
- Free undamped motion
- Free damped motion
- Forced motion
- Undamped forcing and resonance
- Practice problems

McMaster University

Mathematics and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03
Introduction to Differential Equations

Instructor: David Earn

Lecture 24
Spring–mass models: free, damped, and forced motion
Friday 6 November 2026

Announcements

- The TA is Illya Kierkosz (`kierkosi@mcmaster.ca`)
  - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: `https://davidearn.github.io/math2c/`
- The course outline is on the Simple Syllabus web site.
- Avenue to Learn provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on `childsmath`.
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
  - `Subject: Math 2C03: confusion about assignment 1`
  - `Subject: Math 2C03: conflict with test window`
- Office hour: Thursdays, 9:30–10:20 am, after class.

Spring–mass models

[Trench §6.1: undamped motion]
[Trench §6.2: damped motion]

IVP for linear model with constant coefficients

Suppose y is a function of time t. The IVP

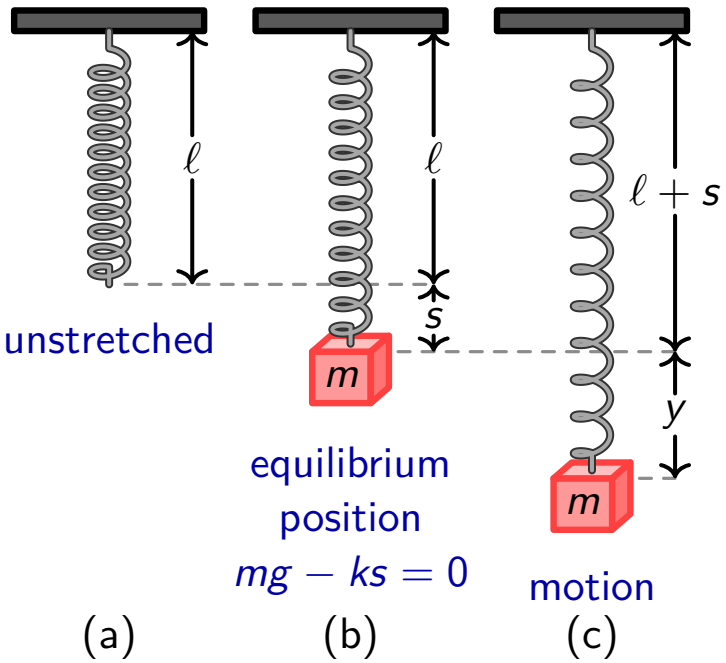
$$a_2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + a_1\frac{\mathrm{d}y}{\mathrm{d}t} + a_0y = g(t)\,, \qquad y(t_0) = y_0,\ \dot{y}(t_0) = y_1,$$

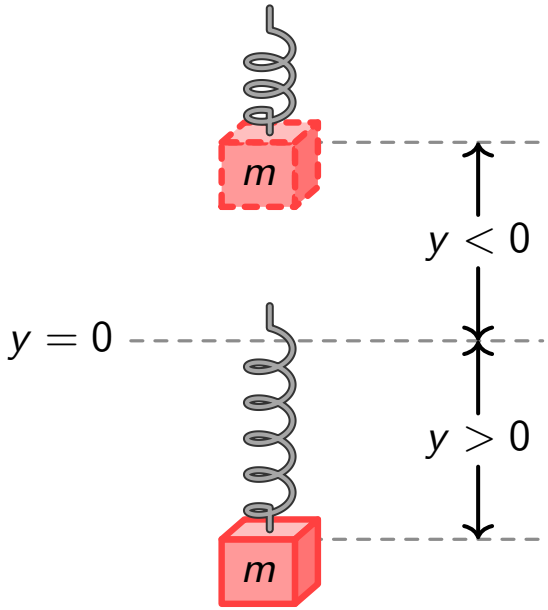
can represent a dynamical system, for which the ***state*** of the system at an initial time t_0 is known, and the state at time t is sought.

Note: To specify the state at time t_0, we need the values of both y and $\dot{y}$ at t_0. Then the future state of the system is determined (according to the existence-uniqueness theorem).

The function $g(t)$ is often called a ***forcing function***, and the solution $y(t)$ (the state of the system) is often called the ***output*** or ***response***.

Hooke's Law





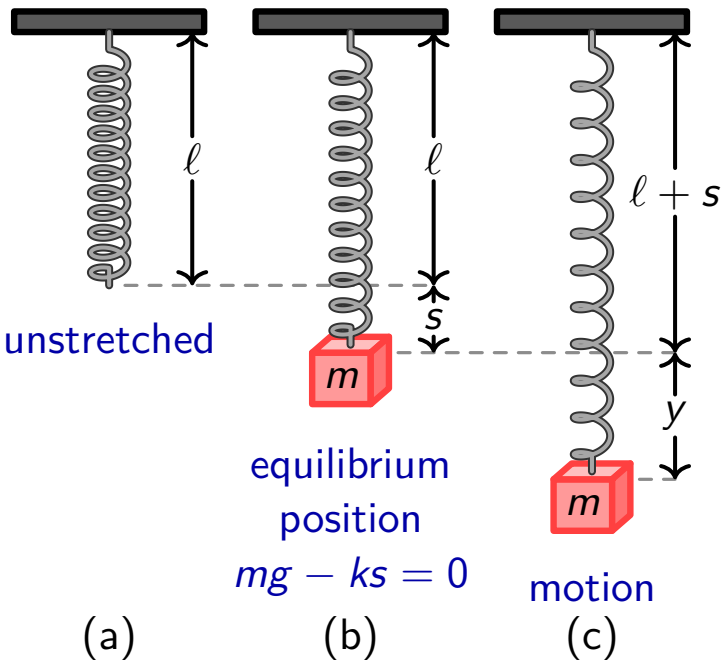
Hooke's Law:

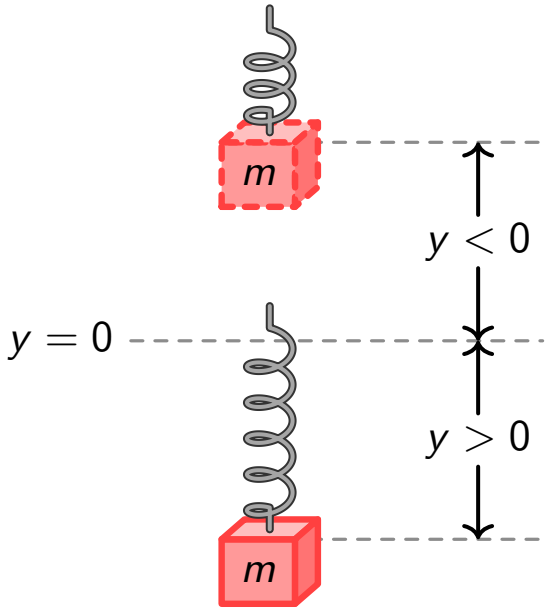
The spring exerts a restoring force F opposite to the direction of stretching, and proportional to the stretch s:

$$F = -ks, \qquad k > 0$$

k is the ***spring constant*** or ***stiffness*** of the spring.

Newton's Second Law





Newton's 2nd Law:

$$m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -k(s+y) + mg = -ky + (mg - ks) = -ky$$

$$\Longrightarrow \quad \frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \omega_0^2 y = 0\,, \qquad \omega_0^2 = \frac{k}{m}.$$

Mass, weight, and units

- *Mass* m and *weight* W are different:
$$W = mg, \qquad m = \frac{W}{g}.$$
- At equilibrium, Hooke's law gives
$$ks = W, \qquad k = \frac{W}{s}.$$
- In SI units, use metres, seconds, kilograms, and newtons:
$$g \approx 9.8\,\mathrm{m/s^2}.$$
- If weight is given in pounds and length in feet, use, *e.g.*,
$$\textit{acceleration}: \quad g = 32\,\mathrm{ft/s^2}, \qquad \textit{mass}: \quad \frac{64\,\mathrm{lb}}{32\,\mathrm{ft/s^2}} = 2\,\mathrm{slugs}.$$
Convert inches to feet before calculating k or initial conditions.

Note: Displacement y is measured positive downward. Trench [§6.1] measures it positive upward, so his displacement has the opposite sign.

Free undamped motion

$$\frac{d^2y}{dt^2} + \omega_0^2 y = 0\,, \qquad \omega_0 = \sqrt{k/m}$$

$\Longrightarrow$ try solution $y = e^{rt}$

$\Longrightarrow$ characteristic equation $r^2 + \omega_0^2 = 0$

$\Longrightarrow$ $r = \pm i\omega_0$

$\Longrightarrow$ general solution $y(t) = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t \qquad (*)$

This ***equation of motion*** (*) describes ***simple harmonic motion*** at ***angular frequency*** ω_0, or ***period*** $T = 2\pi/\omega_0$. The ***frequency*** of motion is $1/T = \omega_0/2\pi$.

Simple harmonic motion is also known as ***free vibration*** (as opposed to ***forced vibration***, which occurs if the equation is non-homogeneous).

Free undamped motion

The equation of motion for simple harmonic motion,

$$y(t) = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t\,, \qquad (*)$$

is more commonly written

$$y(t) = R \cos\left(\omega_0 t - \phi\right), \qquad (**)$$

where the ***amplitude*** R and ***phase*** ϕ are

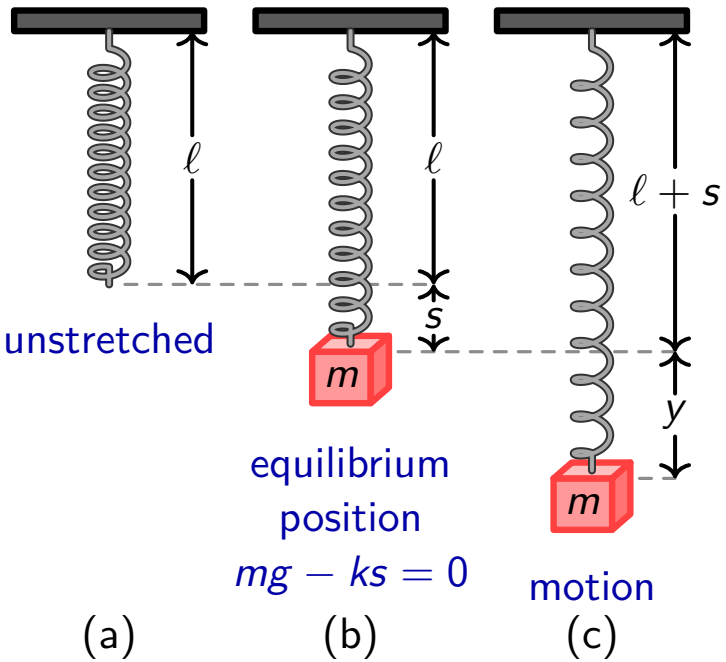
$$R = \sqrt{c_1^2 + c_2^2} \qquad \text{and} \qquad \begin{aligned} R\cos\phi &= c_1, \\ R\sin\phi &= c_2. \end{aligned}$$

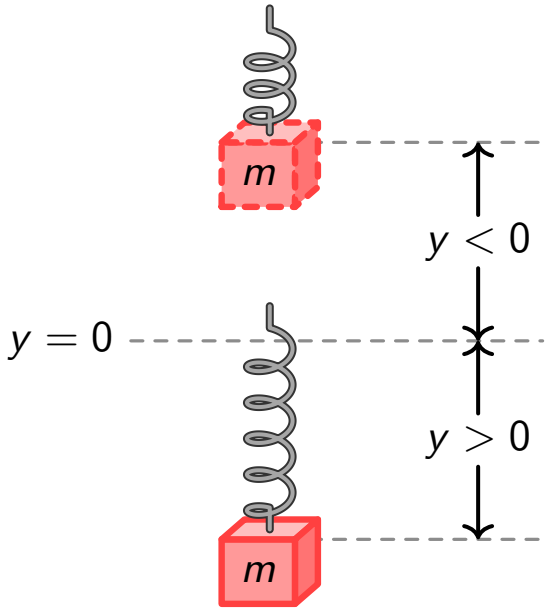
This can be verified using the cosine sum formula:

$$\begin{aligned} R\cos\left(\omega_0 t - \phi\right) &= R\big(\cos\left(\omega_0 t\right)\cos\left(-\phi\right) - \sin\left(\omega_0 t\right)\sin\left(-\phi\right)\big) \\ &= \left(R\cos\phi\right)\cos\omega_0 t + \left(R\sin\phi\right)\sin\omega_0 t \end{aligned}$$

Hence, setting $R\cos\phi = c_1$ and $R\sin\phi = c_2$, we have $R^2\cos^2\phi + R^2\sin^2\phi = R^2 = c_1^2 + c_2^2$ and $\tan\phi = c_2/c_1$ if $c_1 \neq 0$.

Free undamped motion





ODE for simple harmonic motion: $$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \omega_0^2 y = 0\,, \quad \omega_0^2 = \frac{k}{m}.$$

Solution (equation of motion): $$y(t) = R\cos\left(\omega_0 t - \phi\right)$$

Free undamped motion

Example (A brick of mass 2 kg is hanging from a spring)

Suppose the unstretched spring has length $= 0.5\,m$, and the brick is at equilibrium when stretching the spring by 49 cm.

Thus, $m = 2\,\text{kg}$, $\ell = 0.5\,\text{m}$, $s = 0.49\,\text{m}$.

At equilibrium, we know $mg - ks = 0$ so
$k = mg/s = (2\,\text{kg})(9.8\,\text{m/s}^2)/(0.49\,\text{m}) = 40\,\text{kg/s}^2$.

$$\omega_0 = \sqrt{k/m} = \sqrt{(40\,\text{kg/s}^2)/(2\,\text{kg})} = \sqrt{20}\,\text{s}^{-1} \approx 4.47\,\text{s}^{-1}$$

$$T = 2\pi/\omega_0 \approx 1.40\,\text{s}$$

Continued...

Free undamped motion

Example (A brick of mass 2 kg is hanging from a spring)

Suppose the brick is tapped, so it is initially moving down at a speed of 10 cm/s. What is its equation of motion?

$y(t) = R\cos(\omega_0 t - \phi) \implies \dot{y}(t) = -R\,\omega_0 \sin(\omega_0 t - \phi)$

Initial conditions: $y(0) = 0$, $\dot{y}(0) = 0.1\,\text{m/s}$.

$y(0) = 0 = R\cos(-\phi) = R\cos\phi \implies R = 0$ or $\phi = \frac{\pi}{2} + n\pi$, $n \in \mathbb{Z}$.

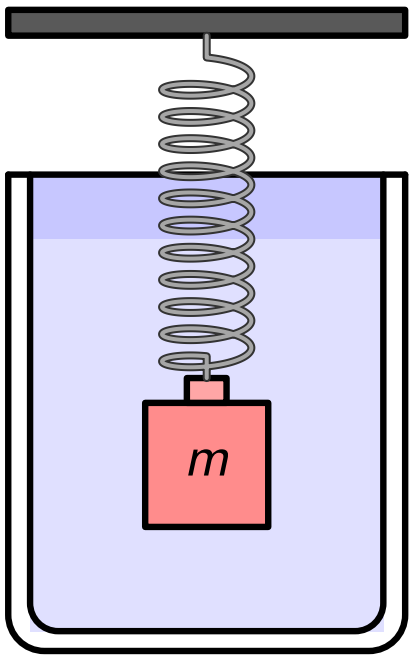
$\dot{y}(0) = 0.1\,\text{m/s} = R\,\omega_0 \sin\phi \implies \phi = \frac{\pi}{2}$ (*i.e.*, $n = 0 \because \dot{y}(0) > 0$)

$\implies R\,\omega_0 = \dot{y}(0)$

$\implies R = (0.1\,\text{m/s})/(4.47\,\text{s}^{-1}) = 0.1/4.47\,\text{m} \approx 0.0224\,\text{m}$

$\therefore R = 2.24\,\text{cm} \implies \boxed{y(t) \approx (2.24\,\text{cm})\cos((4.47\,\text{s}^{-1})t - 1.57)}$

Free damped motion



Resistance *damps* oscillations

$$m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -ky - c\frac{\mathrm{d}y}{\mathrm{d}t}$$

$c > 0$ is the *damping constant*.

Standard form:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \frac{c}{m}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{k}{m}y = 0$$

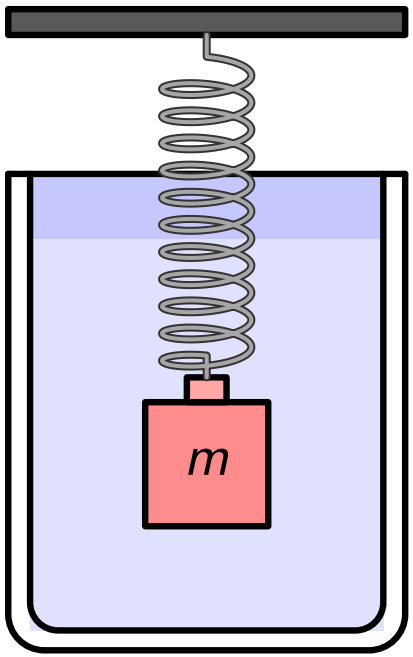
Standard form with convenient parameters:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\lambda\frac{\mathrm{d}y}{\mathrm{d}t} + \omega_0^2 y = 0$$

$$2\lambda = c/m, \quad \omega_0 = \sqrt{k/m}$$

Free damped motion

$$\frac{d^2y}{dt^2} + 2\lambda\frac{dy}{dt} + \omega_0^2\, y = 0$$



characteristic equation: $r^2 + 2\lambda r + \omega_0^2 = 0$

$$\Longrightarrow \quad r = -\lambda \pm \sqrt{\lambda^2 - \omega_0^2}$$

Case (i): $\lambda^2 - \omega_0^2 > 0$
(***overdamped***)

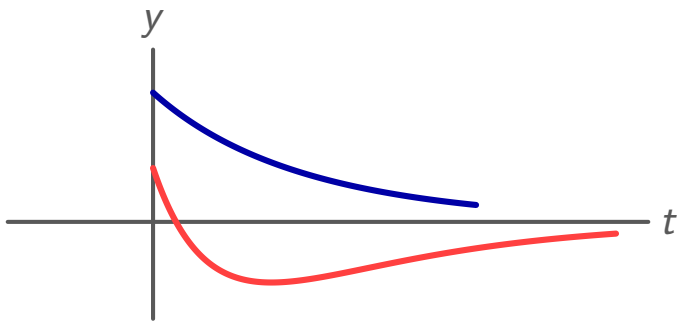
Case (ii): $\lambda^2 - \omega_0^2 = 0$
(***critically damped***)

Case (iii): $\lambda^2 - \omega_0^2 < 0$
(***underdamped***)

Free damped motion

Case (i): $\lambda^2 - \omega_0^2 > 0$ (***overdamped***)

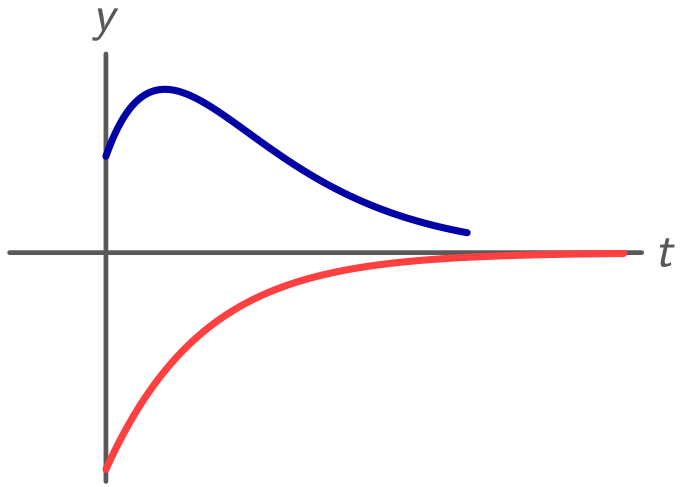
$$y(t) = e^{-\lambda t}\left(c_1 e^{\sqrt{\lambda^2-\omega_0^2}\,t} + c_2 e^{-\sqrt{\lambda^2-\omega_0^2}\,t}\right)$$



Note: $\lim_{t\to\infty} y(t) = 0.$

Case (ii): $\lambda^2 - \omega_0^2 = 0$ (***critically damped***)

$$y(t) = e^{-\lambda t}(c_1 + c_2\, t)$$



Note: $\lim_{t\to\infty} y(t) = 0.$

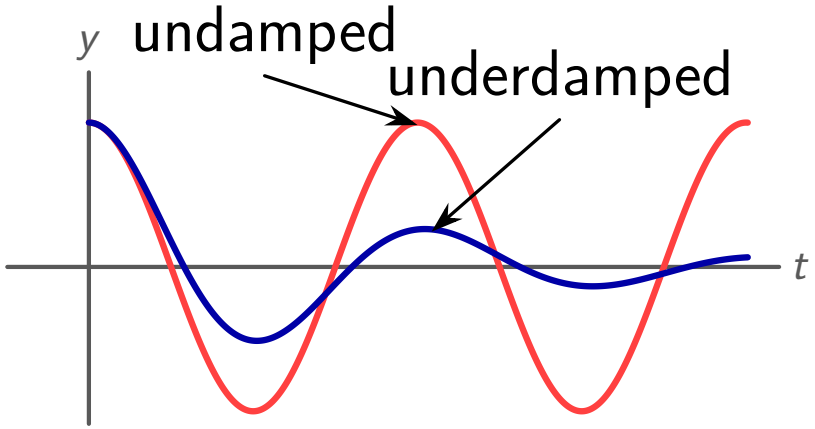
Free damped motion

Case (iii): $\lambda^2 - \omega_0^2 < 0$ (***underdamped***)

Let $\omega_1 = \sqrt{\omega_0^2 - \lambda^2}$.

$$y(t) = e^{-\lambda t}(c_1 \cos(\omega_1 t) + c_2 \sin(\omega_1 t))$$

$$= Re^{-\lambda t} \cos((\omega_1 t) - \phi)$$



Note: $\lim_{t \to \infty} y(t) = 0$.

- Recall $2\lambda = c/m > 0$. For any $\lambda > 0$, *i.e.*, any amount of damping, $y(t) \to 0$ as $t \to \infty$, regardless of value of ω_0.
- Hence, any amount of damping makes $y = 0$ a *stable equilibrium*: initial displacements from $y = 0$ always decay (they "*damp out*").

Equilibrium crossings and extreme displacement

Example ($y'' + 3y' + 2y = 0, \quad y(0) = -1, \quad y'(0) = 5$)

The characteristic equation is

$$(r+1)(r+2) = 0 \quad \Longrightarrow \quad y(t) = c_1 e^{-t} + c_2 e^{-2t}.$$

The initial conditions give

$$c_1 + c_2 = -1, \quad -c_1 - 2c_2 = 5 \quad \Longrightarrow \quad c_1 = 3, \quad c_2 = -4.$$

Thus $y(t) = 3e^{-t} - 4e^{-2t}$.

- *Equilibrium crossing:*

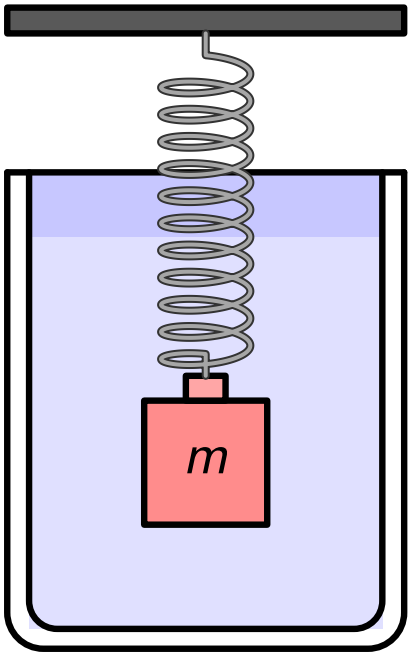
$$y(t) = 0 \quad \Longrightarrow \quad 3e^{t} - 4 = 0 \quad \Longrightarrow \quad \boxed{t = \ln(4/3)}$$

- *Extreme displacement:*

$$y'(t) = -3e^{-t} + 8e^{-2t} = 0 \quad \Longrightarrow \quad \boxed{t = \ln(8/3)}$$

This is a maximum: y' changes from positive to negative.

Forced damped motion



Suppose we apply an ***external force*** $F(t)$ to the brick, *e.g.*, we could push or pull the brick in some way.

$$m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -k\,y - c\frac{\mathrm{d}y}{\mathrm{d}t} + F(t)$$

Standard form:

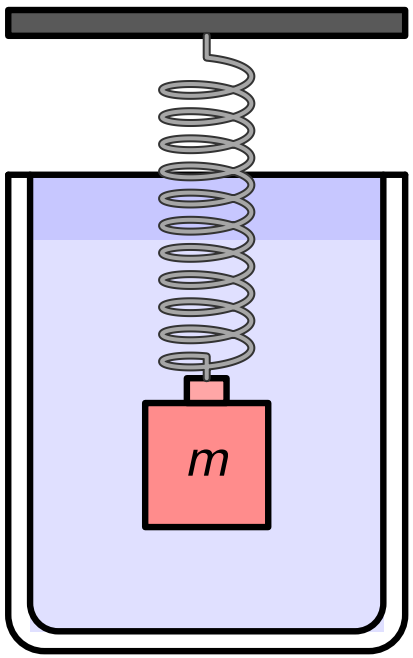
$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \frac{c}{m}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{k}{m}\,y = \frac{F(t)}{m}$$

Standard form with convenient parameters:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\lambda\frac{\mathrm{d}y}{\mathrm{d}t} + \omega_0^2\,y = \frac{F(t)}{m}$$

$$2\lambda = c/m, \quad \omega_0 = \sqrt{k/m}$$

Periodically forced oscillator



Suppose the system is underdamped, so it undergoes *damped oscillations*.

Now suppose we attach a motor that pushes or pulls the brick periodically, such that the external force is $F(t) = F_0 \sin \omega t$.

Dividing by m, we have

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + 2\lambda \frac{\mathrm{d}y}{\mathrm{d}t} + \omega_0^2 \, y = \frac{F_0}{m} \sin \omega t$$

What types of motion are possible for this periodically forced oscillator?

Periodically forced oscillator

$$L[y] = \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + 2\lambda \frac{\mathrm{d}y}{\mathrm{d}t} + \omega_0^2 y = \frac{F_0}{m} \sin \omega t$$

The general solution of this equation can be written $y(t) = y_c(t) + y_p(t)$, where:

- $y_c(t)$ is the general solution to $L[y] = 0$, which we know decays because it contains a factor $e^{-\lambda t}$;
- $y_p(t)$ is the particular solution that does not decay with time;
  - any other particular solution contains terms from $y_c(t)$, which decay;
  - find $y_p(t)$ via undetermined coefficients;

$y_c(t)$ specifies the ***transient dynamics***, whereas $y_p(t)$ specifies the ***asymptotic dynamics*** (what remains after the transient behaviour has damped out).

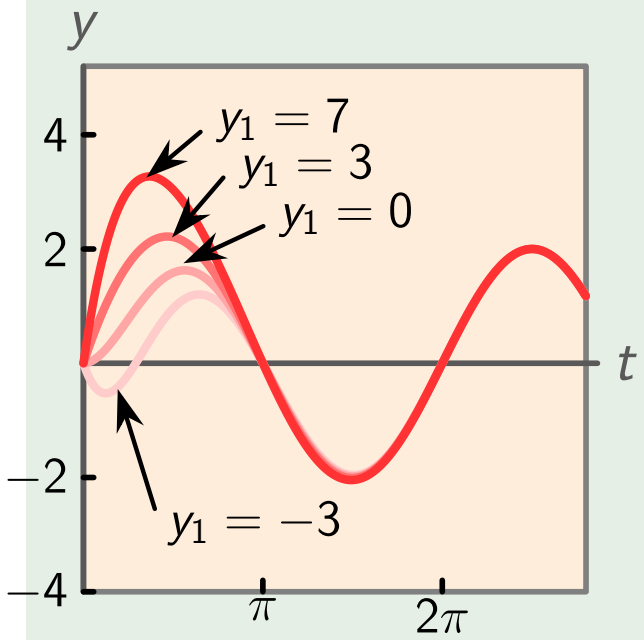
Note: Trench [after Eq. (6.2.17)] uses the term ***steady state component*** for the asymptotic dynamics.

Periodically forced oscillator

Example (Transient and steady state solutions)

IVP: $$\frac{d^2y}{dt^2}+2\frac{dy}{dt}+2y = 4\cos t+2\sin t\,, \quad y(0)=0,\ y'(0)=y_1.$$

Solution: $$y(t) = \underbrace{(y_1-2)}_{\text{constant}}\underbrace{e^{-t}\sin t}_{\text{transient}} + \underbrace{2\sin t}_{\text{steady state}}$$



- all solutions start at equilibrium ($y = 0$);
- different solutions have different initial velocities (y_1);
- transient dynamics damp out quickly;
- all solutions quickly converge onto the asymptotic dynamics (steady state solution).

Periodically forced oscillator

Example (Undamped forced motion)

IVP: $$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \omega_0^2 y = \frac{F_0}{m}\sin\omega t\,, \quad y(0) = 0,\ y'(0) = 0, \quad \omega \neq \omega_0.$$

Complementary function:

$$y_c(t) \;=\; R\cos\left(\omega_0 t - \phi\right), \qquad R \geq 0,\ \phi \in [0, 2\pi).$$

Particular solution via undetermined coefficients:

$$y_p(t) \;=\; \frac{F_0}{m(\omega_0^2 - \omega^2)}\sin\omega t\,.$$

General solution:

$$y(t) \;=\; R\cos\left(\omega_0 t - \phi\right) + \frac{F_0}{m(\omega_0^2 - \omega^2)}\sin\omega t\,.$$

Periodically forced oscillator

Example (Undamped forced motion)

$$y(t) = R\cos(\omega_0 t - \phi) + \frac{F_0}{m(\omega_0^2 - \omega^2)}\sin\omega t$$

$$y'(t) = -R\,\omega_0 \sin(\omega_0 t - \phi) + \frac{\omega F_0}{m(\omega_0^2 - \omega^2)}\cos\omega t\,.$$

The initial conditions give

$$R\cos\phi = 0, \qquad R\omega_0 \sin\phi + \frac{\omega F_0}{m(\omega_0^2 - \omega^2)} = 0$$

$$\text{Hence} \quad R\cos\phi = 0, \qquad R\sin\phi = -\frac{\omega F_0}{m\omega_0(\omega_0^2 - \omega^2)}.$$

Using amplitude–phase form to expand the complementary term,

$$y(t) = R\cos\phi\cos\omega_0 t + R\sin\phi\sin\omega_0 t + \frac{F_0}{m(\omega_0^2 - \omega^2)}\sin\omega t$$

$$= \frac{F_0}{m(\omega_0^2 - \omega^2)}\left(-\frac{\omega}{\omega_0}\sin\omega_0 t + \sin\omega t\right).$$

We assumed $\omega \neq \omega_0$. *What happens if $\omega = \omega_0$?*

Periodically forced oscillator

Example (Undamped forced motion)

To solve, either start over with $\omega = \omega_0$ or consider the limit $\omega \to \omega_0$:

$$\lim_{\omega\to\omega_0} y(t) = \lim_{\omega\to\omega_0} \left[\frac{F_0}{m(\omega_0^2 - \omega^2)} \left(-\frac{\omega}{\omega_0} \sin(\omega_0 t) + \sin \omega t \right) \right]$$

This limit has form $\frac{0}{0}$. *Use L'Hôpital's rule.*

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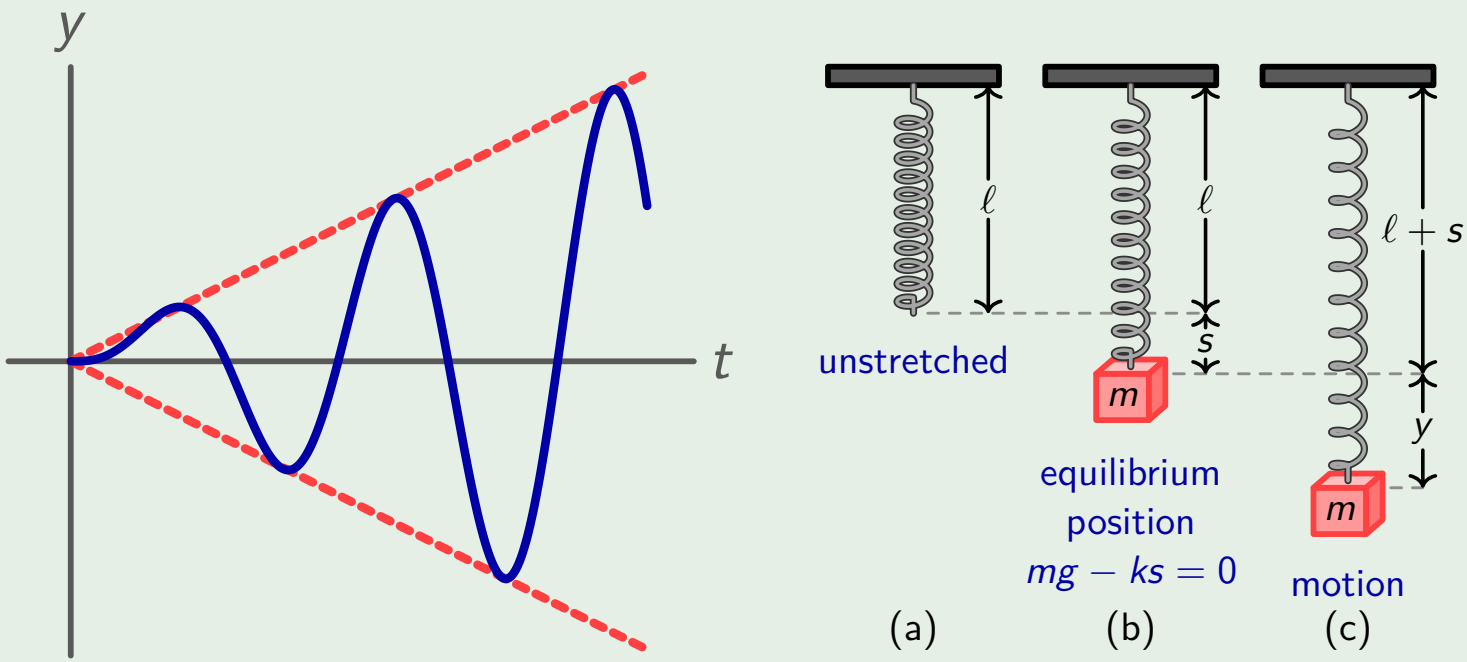
$$y(t) = \frac{F_0}{2m\omega_0^2} \sin \omega_0 t - \frac{F_0}{2m\omega_0} t \cos \omega_0 t$$

Amplitude $R = \sqrt{\left(\frac{F_0}{2m\omega_0^2}\right)^2 + \left(\frac{F_0}{2m\omega_0} t\right)^2}$

Amplitude grows with time t: ***resonance***.

Periodically forced oscillator

Example (Undamped forced motion)



Does this make sense physically for arbitrarily large t? Please do Spring resonance.

Can the amplitude exceed the spring length ℓ? Can we have $|y| > \ell + s$? For large enough amplitudes of oscillation, modelling the system with a linear ODE is inadequate. We must consider nonlinearity in the restoring force, or some other nonlinear effect(s).

Poll

- Go to
  https://www.childsmath.ca/childsa/forms/main_login.php
- Click on Math 2C03
- Click on Take Class Poll
- Fill in poll **Spring resonance**
- Submit.

A decaying force need not leave a motionless mass

Example ($y'' + 4y = 10e^{-t}, \quad y(0) = y'(0) = 0$)

The complementary solution is $y_c = c_1 \cos 2t + c_2 \sin 2t$. Try $y_p = Ae^{-t}$:

$$y_p'' + 4y_p = 5Ae^{-t} = 10e^{-t} \quad \Longrightarrow \quad A = 2.$$

Thus $y = c_1 \cos 2t + c_2 \sin 2t + 2e^{-t}$. The initial conditions give

$$c_1 + 2 = 0, \qquad 2c_2 - 2 = 0, \quad \Longrightarrow \quad c_1 = -2, \quad c_2 = 1.$$

Therefore

$$\boxed{y(t) = -2\cos 2t + \sin 2t + 2e^{-t}}$$

As $t \to \infty$, the last term decays, leaving an oscillation with amplitude $R = \sqrt{(-2)^2 + 1^2} = \sqrt{5}$. There is no damping to remove this remaining motion.

Trench practice problems

- [Trench §6.1: undamped spring motion]
  - Free motion and initial conditions: 1, 3, 4, 6.
  - Forced motion and resonance: 9, 11, 15.
- [Trench §6.2: damped spring motion]
  - Free damped motion: 1, 3, 7, 9.
  - Forced motion and its steady state component: 17, 20.
  - Equilibrium crossings: 22, 23.

Remember: Trench [§6.1] measures displacement positive upward. Translate the initial conditions and the sign of any external force consistently if using our downward-positive convention.